

Reference architecture for the digitization of industries

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Summary and further research needs

The International Data Space (IDS) offers an information technology architecture for safeguarding data sovereignty within the corporate ecosystem. It provides a virtual space for data where data remains with the data owner until it is needed by a trusted business partner. When the data is shared, terms of use can be linked to the data itself.

Analysis of six use cases from the first phase of the prototype implementation of the IDS architecture shows that the focus lies on the standardized interface, the information model for describing data assets, and the connector component. Further use cases are planned for the next wave of implementation that are based on the broker functionality and require the use of vocabularies for simple data integration.

In addition, companies need to standardize the principles that are translated into the terms of use. These principles need to be shaped, described, documented, and implemented in a simple and understandable way. They also need to be understood in the same way by different actors in the corporate ecosystem, thus requiring semantic standardization.

Furthermore, the IDS Reference Architecture Model needs to be set in context with respect to related models. In the F3 use case, an OPC UA adapter is used. Additional use cases for integration with the Plattform Industrie 4.0 administration shell and Industrial Internet Reference Architecture are pending.

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The IDS Architecture is also increasingly being utilized in so-called verticalization initiatives, in healthcare and in the energy sector for example. These kinds of initiatives – like the Materials Data Space – demonstrate the cross-domain applicability of the architectural components and provide information about further development needs.

Finally, in anticipation of the future development of the use cases and utilization of the IDS, work on the economic valuation of data and on the settlement and pricing of data transactions must be accelerated.

8.1 Introduction: digitization of industry and the role of data

The digitization of society and economy is the major trend of our time. The introduction of consumer technologies into businesses, the shift towards digital services for companies' business offerings, and the networking of things, people and data open up new business opportunities in nearly all sectors of the economy. Many of these have already been analyzed by the Smart Service Welt ("Smart Service World") working group within Plattform Industrie 4.0. There are five distinguishing features of the smart service world [1].

- End-to-end customer process: smart services do not address particular customer needs but rather support an entire customer process. Sensor manufacturer Endress+Hauser, for example, does therefore not only offer measuring systems but supports customers in all of their questions from planning a production facility, from advice on product selection, installation and commissioning, and configuration during operation, to advice on replacement investments and plant replacement [4].
- Hybrid service bundles: are no longer just the products central to the product offering but bundles of physical products and digital services. An example would be the sportswear manufacturer adidas, which, in addition to running shoes, also offers the "runtastic" digital service in order to support its customers' running experience – the customer process – in its entirety.
- Data at the center: hybrid service bundles, which support the end-to-end customer process, are only possible, if companies efficiently and effectively link their own data with the data of business partners and contextual data – maintaining data sovereignty above all. This contextual data often stems from public data sources and is available as open data. Examples include location data, traffic information and so on.

- Business ecosystems: the need to combine data from various sources in order to support the end-to-end customer process leads to the development of business ecosystems. These are networks of different actors, which form dynamically around an end-to-end customer process [5]. Well-known examples are the Apple App Store and Uber.
- Digital platforms: the interaction within the ecosystem and the sharing of a broad variety of data in the interest of the end-to-end customer process require new information systems. Digital platforms provide this functionality according to the “as a service” principle.

The overall characteristics of the smart service world facilitate new business models, both for new and established companies. For the latter, the key element in digitization is not simply copying new digital business models, but rather supplementing existing strengths – often in product design, production or sales – with smart service world opportunities. One example is preventative maintenance services by plant manufacturers that draw on usage data from a number of plant operators for the early identification of maintenance needs, thus improving the overall equipment efficiency (OEE).

For the manufacturing industry in particular, digitization changes not only the product offering but also the product manufacturing side. This is because the consequence of the hybrid service bundle on the offering side is an increase in product and above all process complexity on the product manufacturing side. Industry 4.0 offers a solution to managing this complexity founded on the organizational princi-

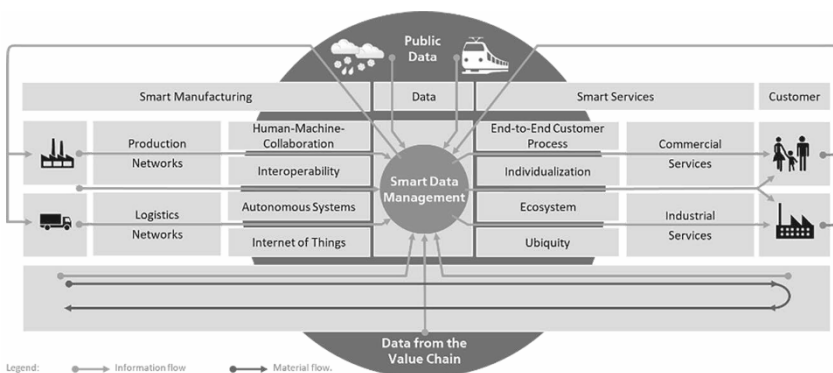


Fig. 8.1 Smart Data Management from the perspective of the focal business (Fraunhofer-Gesellschaft/Industrial Data Space)

ples of autonomy, networking, information transparency and assistance capability of systems for product manufacturing[3][2].

Both the product manufacturing and the product offering sides are linked via a modern data management system (see Fig. 8.1).

The following basic assumptions are applicable when designing this smart data management system:

- Data is a strategic and economically valuable resource.
- Companies need to share data more intensively than in the past. This applies to the efficiency, effectiveness, and especially flexibility of conventional data sharing as well as to the extent of the data to be shared. More and more, companies need to incorporate data into the business ecosystem that was previously classified as too sensitive to be made accessible to third parties.
- Companies only enter into this data sharing, if their data sovereignty is guaranteed, that is, if they can decide who is allowed to use their data on which conditions and for what purpose when the data leaves the company.

Data sovereignty is thus the capacity of exclusive self-determination of a natural person or corporate entity with respect to the asset of data [7]. Data sovereignty is expressed in the balance between the need to protect data and its shared use in the business ecosystem. It is a key competency for success in the data economy.

Alongside economic and legal frameworks, it is essential to create the information technology conditions so that data sovereignty can be exercised. This is why the IDS initiative was brought into being.

8.2 International Data Spaces

8.2.1 Requirements and aims

The IDS initiative pursues the goal of digital sovereignty in the business ecosystem. The IDS participants' aim is to be able to share data with business partners interoperably while always retaining self-determination with regards to these data assets.

This aim is to be achieved by designing an IT architecture, demonstrating its applicability and usefulness in use-case projects. The initiative is founded on two pillars: a Fraunhofer research project and the International Data Spaces Association (IDSA). The research project is being carried out by Fraunhofer funded by the Federal Ministry of Education and Research. Both the project and the association

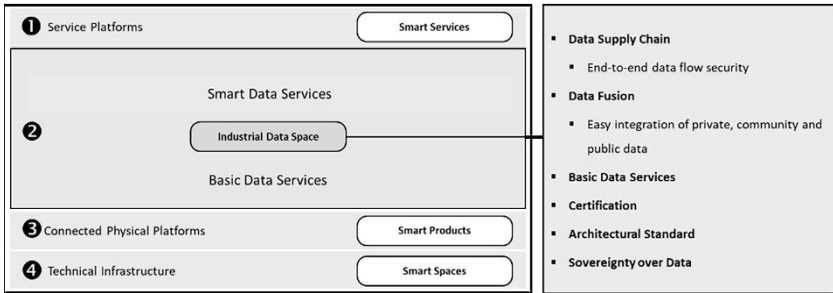


Fig. 8.2 Smart service world requirements for the International Data Space (Smart Service World working group)

are precompetitive and non-profit. The initiative encourages the broadest possible dissemination and thus also welcomes commercial paths of exploitation, which are open to all market participants.

From the overarching goal of facilitating confident data sharing in business ecosystems (smart service world), we are able to develop key requirements for the architecture of the IDS (see Fig. 8.2) [8]:

- A1 Terms of use for data: when the data is shared, the data owner is able to link enforceable terms of use to the data – rules – that specify on which conditions the data may be used, by whom, and for what purpose.
- A2 Secure data supply chains: the entire data value chain, from the creation/generation of the data (e.g. via sensors) to its use (e.g. in smart services) must be able to be secured.
- A3 Simple data linking: in-house data must be able to be easily linked with that of business partners but also with (public) contextual data.
- A4 Decentralized data storage: the architecture must not necessarily require central data storage¹. Rather, it must be possible to only share the data with trusted partners if it is required from a clearly identifiable partner in keeping with the terms of use.
- A5 Multiple operating environments: the software components of the IDS Architecture, which facilitate participation in this virtual data space need to be able to be run in conventional company IT environments, but also, for example, on IoT cloud platforms and mobile devices or sensor PCs.

¹ Two thirds of companies mistrust central *data lake* architectures, for example, because they fear that third parties will have unwanted access to the data cf.[10].

- A6 Standardized interface: data sharing in the IDS must take place in accordance with a pre-defined and open information model.
- A7 Certification: software components and participants must be certified with respect to keeping to the requirements of Industrial Data Space software and its operation. The certification criteria are the responsibility of the International Data Spaces Association.
- A8 Data apps and app store: data services (data apps) provide essential functions based on the reference architecture for handling data in the IDS. Examples are data format conversion and assigning terms of use to the data. These data services need to be made available via an app store functionality.

These requirements guide the development of the Reference Architecture Model in the IDS.

8.2.2 International Data Space Reference Architecture Model

The Reference Architecture Model defines the International Data Space's individual components and their interaction [9] in terminologically and conceptually consistent terms. It distinguishes five levels and three perspectives.

The levels are:

- The business level describes the roles in the IDS.
- The functional level describes, in technological- and application-agnostic terms, the specialist and functional requirements for the IDS.
- The process level describes the interactions between the roles and the associated specialist functions.
- The information level describes the entities within the IDS and their relationships to one another in domain-independent terms.
- The system level describes the software components of the IDS.

The three perspectives are *security*, *certification* and *data governance*. They run perpendicular to the levels.

Fig. 8.3 depicts the model of roles in the IDS as part of the business level.

Data Owner, Data Provider and Data User are the core roles in the IDS. Data sharing between these core roles is supported by the following intermediary roles:

- Broker: enables data sources to be published and found.

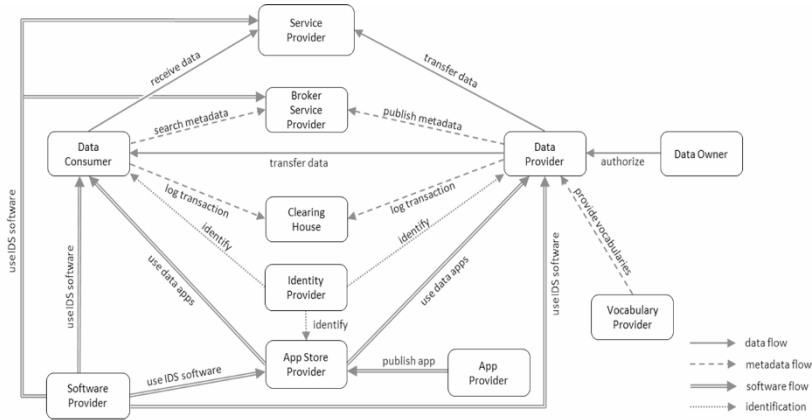


Fig. 8.3 Model of roles in the IDS (Fraunhofer-Gesellschaft/International Data Spaces Association)

- Clearing house: logs data sharing processes and resolves conflicts.
- Identity provider: issues digital identities/certificates.
- Vocabulary provider: provides semantic information models to, for example, specific domains².

Software and services providers offer the software and services necessary for fulfilling the different roles.

Finally, the certification authority and one or more inspection bodies ensure that all Industrial Data space requirements are fulfilled by the software and the participants.

8.2.3 State of development

The work is organized as a consortium-based research project³ with researchers from a total of twelve Fraunhofer institutes working alongside representatives of the companies within the International Data Spaces Association. The project follows a design-based research approach where the Reference Architecture Model is implemented in software prototypes in agile development sprints. The companies use the

² The IDS draws on VoCol technologies, software that supports the shared creation and management of vocabularies [9].

³ For details on consortium-based research cf. [6].

prototypes in their use cases in order to demonstrate the feasibility and applicability of the architecture and the associated benefits, as well as identifying additional development needs.

Twenty months into the project, six software components are available as prototypes:

- K1 Connector: The central component is the IDS Connector that is used by the core roles. Prototypes are available for a basic version (without functionality for exercising usage control), for a so-called Trusted Connector (with usage control), for a sensor connector, and for an embedded version (for mobile end devices, too).
- K2 Usage control: A version of the INDUCE framework [11] and processes for data labeling are available as technology packages and are additionally integrated into the Trusted Connector.

Table 8.1 Quality function illustration for the IDS

		K1				K2	K3	K4	K5	K6
		Connecto				Usage Control	Information Model	App Store	Base Broker	Identity Provider
		Base Connector	Trusted Connector	Sensor Connector	Embedded Connector					
A1	Terms of use for data		X	(X)	(X)	X	X			
A2	Secure data supply chain			X			X			X
A3	Simple data linking						X		X	
A4	Decentralized data storage	X	X	X	X		X		X	
A5	Multiple operating environments	X	X	X	X					
A6	Standardized interface	X	X	X	X		X	(X)	X	
A7	Certification						X			
A8	Data apps and app store							X		

Key: X – Requirement fulfilled; (X) – requirement partially fulfilled

- K3 Information model: The model has not only been conceptualized, it is also available as a software library and can thus be used by software developers within the International Data Spaces Association.
- K4 App store: An initial prototype is available.
- K5 Base broker: An initial prototype of the broker with basic registry functionality is available.
- K6 Identity provider: An initial prototype version of an identity provider service is available.

Table 8.1 shows to what extent the requirements for the IDS are fulfilled by six software components.

All of the requirements are fulfilled by at least one component. Thus, the requirement for the terms of use is addressed both conceptually in the information model and implemented in initial versions, in particular as the Trusted Connector. The requirement for various operating environments is comprehensively implemented via four versions of the Connector architecture. However, requirements A7 (Certification) and A8 (Data apps and app store) are only addressed in initial basic components.

8.3 Case studies on the International Data Space

8.3.1 Collaborative supply chain management in the automotive industry

A growing number of models, derivatives, in-car functions, and shorter product lifecycles are leading to increasing supply chain complexity in the automotive industry. Modern mid-size vehicles offer so many options and versions that theoretically more than 10^{30} varieties of configuration are possible. Around three quarters of the components required for this do not originate from the manufacturer (original equipment manufacturers, OEMs) but from suppliers. This product complexity can only be efficiently and effectively replicated in production and logistics processes, if suppliers and OEMs work closely together, not only at the planning stage (of requirements, capacities and job sequencing) but also during the execution of the processes (during transport, manufacture, and assembly). In addition to data that has been shared for decades through electronic data interchange (EDI) solutions, more and more data must be shared today that was considered too sensitive to be shared in the past. This includes:

- Inventory days for specific critical components
- Detailed information on manufacturing steps for critical components in the supplier network
- Structure of the supplier network
- Added value information for component transport (heat, shocks and vibrations, etc.)

This data will only be shared, if the data owner can specify the terms of use. This is where the IDS comes into play.

Fig. 8.4 shows the software systems and data flows in the use case of collaborative supply chain management in the automotive industry. In the first phase, the use case encompasses a tier 1 supplier and an OEM. Both use the Base Connector to support the following data sharing operations during the project's initial phase:

In step one, the tier 1 supplier informs the OEM of a supply risk for one of his suppliers (tier 2) for a subcomponent that the OEM needs. To do this, the tier 1 supplier combines data from his risk management system with master records from the supplier system and transmits this data to the OEM via the IDS Connector, including the terms of use that this data is only to be used for a specific supplier management purpose for a period of 14 days.

The OEM imports this data into its supplier management system and uses it to calculate the updated inventory days for specific components that it obtains from the tier 1 supplier. The OEM in turn sends this inventory days data via the IDS

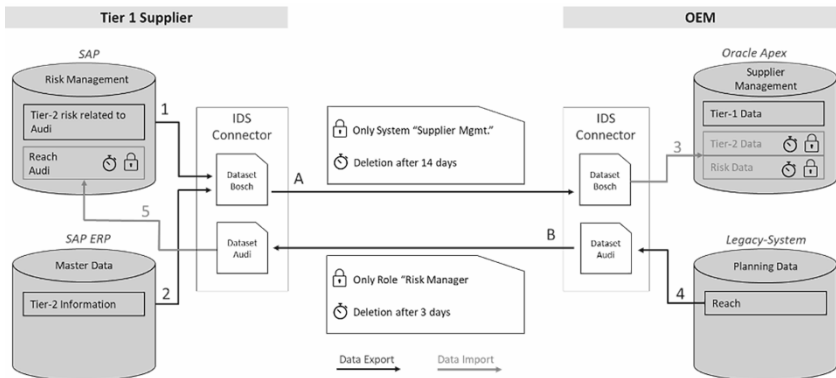


Fig. 8.4 System components in collaborative supply chain management (Chair for Industrial Information Management TU Dortmund University - formerly Audi Endowed Chair Supply Net Order Management TU Dortmund University)

Connector to the tier 1 supplier, again including the terms of use, stating the purpose for use (risk management) and the maximum days of use (three days).

The benefit for the OEM of this use case is obvious. Supply risks in production, “breaks in the supply chain”, are recognized sooner and manufacturing downtime therefore avoided. For the tier 1 suppliers, the benefit lies principally in improved ability to plan their own production since the OEM’s inventory days are made available to them.

This use case is currently in the software prototype implementation phase.

8.3.2 Transparency in steel industry supply chains

Steel production is a transport intensive business, with individual truck shipments being subject to interruption generally due to delays during transportation (traffic on the main leg on the freeway, situation at the factory gate, etc.).

All of the stakeholders in the supplier network – in particular the supplier and the logistics service provider as well as the individual transportation company, the producing company, additional logistics service providers for distributing the fin-

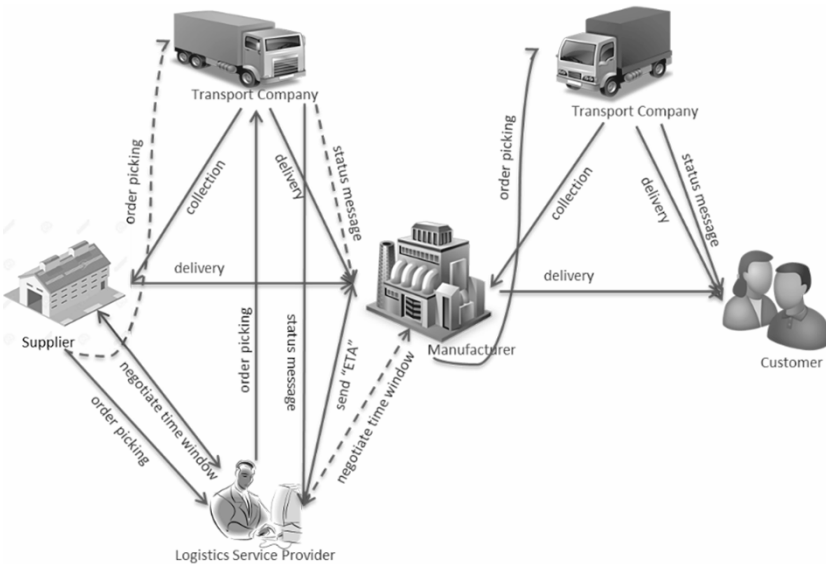


Fig. 8.5 Transparency in steel industry supply chains (Fraunhofer ISST)

ished products and the end customer themselves – have an interest in being informed in real time about events in the supply chain that lead to plan changes.

This use case addresses notification of delayed arrival on the inbound side of the manufacturing company. Here, the transportation company informs the logistics service provider, using a mobile version of the IDS Connector, that a specific shipment is delayed and states the reason for this delay. The logistics service provider calculates an updated expected arrival time and transmits this to the manufacturing company awaiting the consignment. The manufacturing company, in turn, confirms the new arrival time and transmits updated details in regards to the unloading point.

Data is shared according to the IDS information model. For the payload itself “packed” in the IDS notification, GS1 EDI XML is used.

The benefit for the manufacturing business lies in the improvement of the planning of inbound logistics processes/yard management (including staff planning within the receiving department) and production (including job controlling). For the logistics service provider, the use case offers value-added services for customers. For the transportation company the check-in process at the manufacturing company is made easier because updated information regarding time slots and unloading points is always available.

This use case is currently being piloted with thyssenkrupp and already covers more than 500 shipments per month.

8.3.3 Data trusteeship for industrial data

The model of roles for the IDS is fundamentally structured so as to allow for a separation of the role of Data Owner from that of Data Provider. This allows for new business models, specifically data trusteeship.

The processing of data across different service and process steps involves significant requirements in terms of data protection and data sovereignty that not all companies render themselves, but obtain as a service. The data trustee thus needs to ensure that data does not leak out to competitor companies, that personal data is adequately anonymized, and that services are instructed to only use data in accordance with the terms, deleting the data where necessary after use.

Independent audit firms and technical testing organizations are well suited to provide data sovereignty services in business ecosystems, including the following:

- Reviewing rules on terms of use for any conflicts
- Monitoring observation of terms of use
- Monitoring data transactions as a clearing house

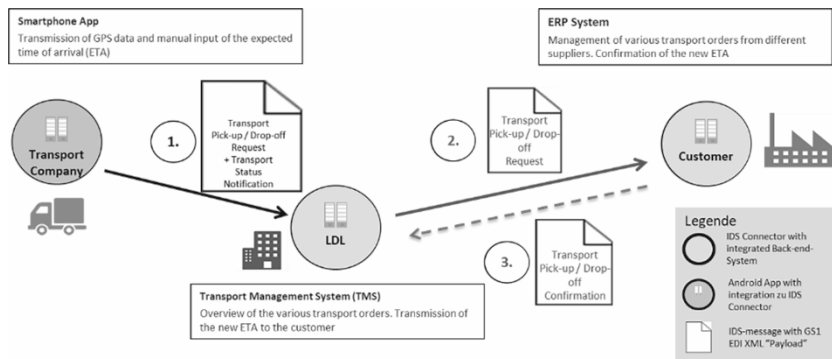


Fig. 8.6 System chain for transparency in the supply chain (Fraunhofer ISST)

- Data processing, enrichment, and provision on behalf of third parties
- Auditing and certification services within the IDS

The certification requirements and criteria, roles and certification levels, and auditing methods in the IDS are defined in a certification schema. Every organization that participates in the IDS is audited and certified according to this schema, as are all of the IDS's software components.

In addition to this, the certification body also facilitates, for example certificate-based identity management within the IDS, which is the technical basis for all Connector implementations.

This use case is currently in the conception phase.

8.3.4 Digital networking of manufacturing lines

Industry 4.0 is an organizational principle for the industrial operations of the future and rests on, among other things, the networking of all resources within product manufacturing. Machines, facilities, and staff within manufacturing are able to share information in near real time, transmitting manufacturing status data and order data, but also receiving contextual information on individual production steps. Since in many industries today manufacturing happens in distributed production networks, companies have two challenges to overcome:

- Shared semantic descriptions of manufacturing resources in the production network
- Confident data sharing between the individual resources (e.g. machines)

The use case in the IDS addresses both challenges.

Linked data principles such as RDF as the *lingua franca* for data integration and the associated W3C standards form the basis for the semantic descriptions. In this way, information architectures that have evolved over time can be transformed into knowledge-based information networks that then form the common informational basis for digital product manufacturing processes and innovative smart services. The data here is structured and semantically enriched in such a way that existing data silos are overcome, and data value chains can be established across functions and processes. One of the use cases for this in the IDS is the development of a knowledge graph for the production of the company Schaeffler. Here, concepts such as the administration shell model from the Industry 4.0 reference architecture model are used alongside the IDS information model. For the connection, data from the source systems (e.g. manufacturing execution systems and sensor data from machines and production) is transformed into RDF vocabularies.

If a shared semantic model – a vocabulary – is established in the production network then data can be confidently shared and its meaning also immediately be understood. The digitization of manufacturing lines use case also provides an OPC-UA adapter that can be run as a data service in the Industrial Data Connector; it facilitates the importing of OPC UA compliant data and its linking with data from other sources (e.g. ERP systems, see Fig. 8.7). In this way, the data is made available for a range of usage scenarios in the production network.

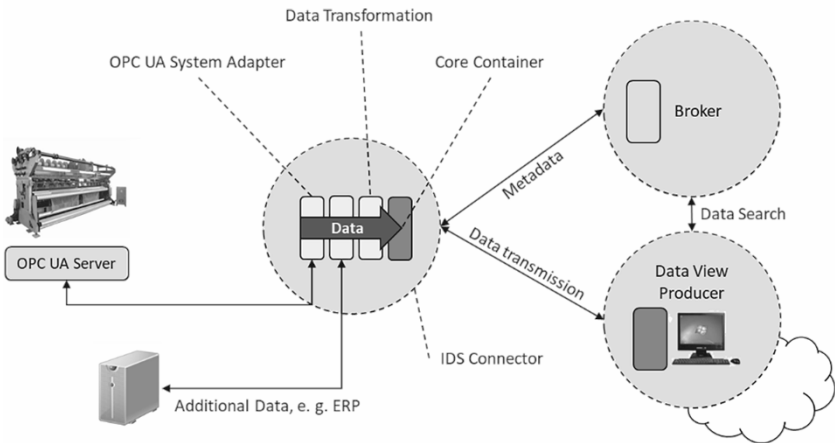


Fig. 8.7 Digital networking of manufacturing lines (Fraunhofer IOSB)

8.3.5 Product lifecycle management in the business ecosystem

More than two thirds of all new products are based on new materials. In order to safeguard the innovational strength and retain technological sovereignty and closed value chains within Germany, universal digitization of material and product properties over their entire lifecycle (“from ore to the fridge”) is of strategic importance.

The Materials Data Space provides digitized information on materials and their properties, components, and their alterations in manufacturing and use over the entire value chain, thus covering their entire lifecycle from use to strategic recycling. The Materials Data Space, a verticalization of the IDS architecture, is a strategic initiative of the Fraunhofer MATERIALS group, which aims for comprehensive digital image within the entire business ecosystem (see Fig. 8.8).

The steel industry, for example, does not only sell steel strips themselves but also the digital image of the steel strip (microstructure, composition, inclusions, material history, etc.). The combination of the physical product together with the digital image is a key success factor for the future competitiveness of companies such as Salzgitter AG, the implementation partner in this use case. In addition, combining the data over the entire lifecycle results in shorter development times and learning

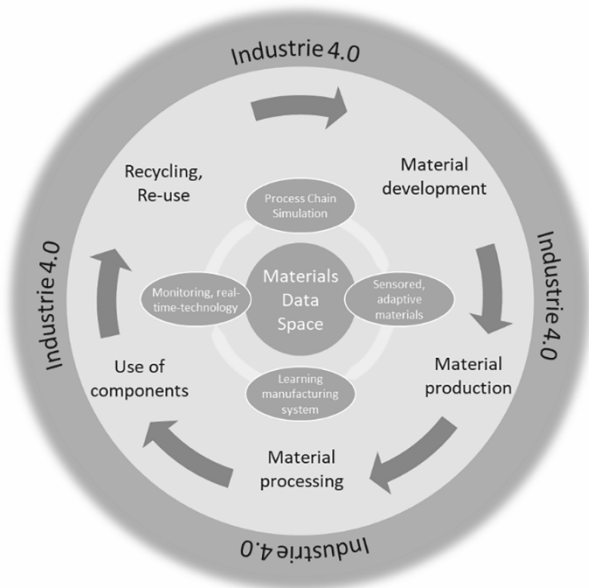


Fig. 8.8 Materials Data Space (Fraunhofer-Gesellschaft)

manufacturing processes. What is more, fundamental potential benefits with respect to materials and production efficiency and recycling accrue for the entire business ecosystem.

This use case is supported by a Fraunhofer preliminary research project identifying domain-specific requirements for IDS components as well as for the terms of use that will be linked with the data of the various actors in the Materials Data Space business ecosystem.

8.3.6 Agile networking in value chains

Networking in value chains is not only limited to companies or people but, with the spread of the Internet of Things, increasingly stretches to objects, too. Load carriers, containers, trucks, etc. are all uniquely identifiable, communicate with their environment, and make decisions independently (with respect to shipping routes, for example). In the course of this, these objects produce data, specifically value chain status data. They provide information, for example, on locations (including time stamps), atmospheric conditions (temperature, humidity), and shocks.

Agile networking in value chains requires both confident sharing of this kind of status data by the objects themselves and the identification of relevant data sources.

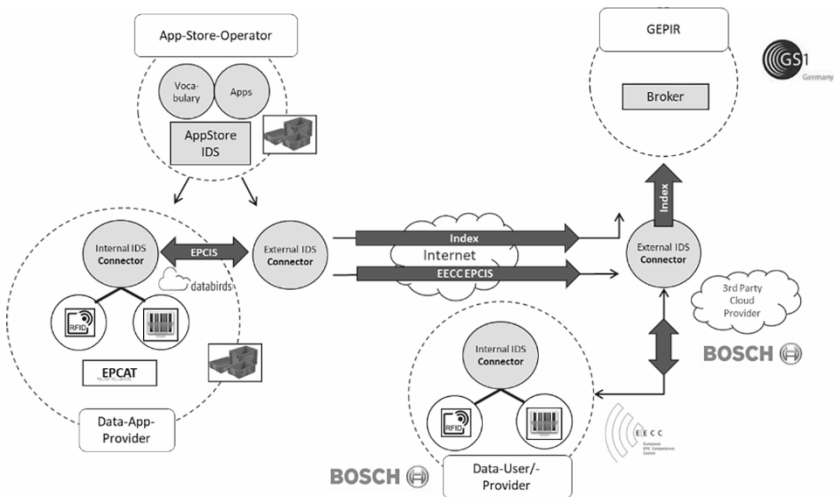


Fig. 8.9 Overall architecture for the STRIKE use case (Fraunhofer-Gesellschaft/International Data Space)

This is because master records and status data from small load carriers are shared in value chains with changing business partners across companies.

In the STRIKE use case (Standardized RTI Tracking Communication Across Industrial Enterprises), status data is sent via the IDS Connector in a useable format for the recipient. This status data is produced and managed using the EPCIS standard. Using the example of new load carriers equipped with RFID, a demonstration is given of how the status data can be assigned to the data owner within the GS1 community. The data owners therefore remain sovereign over their data.

The existing GS1 partner registry, GEPIR (Global Electronic Party Information Registry), is also being developed into an Industrial Data Space broker in order to be able to identify providers of status data in the value chain. In addition, this example shows how value-added services can be derived from the data with the help of apps.

Fig. 8.9 shows the overall architecture for the STRIKE use case.

8.4 Case study assessment

Use cases assessment mirrors the strategy for the implementation of the prototype reference architecture model (see Table 8.2). This is because prototypes of the IDS Connector, as the architecture's core component in various implementations, are being put to work or have been planned in all of the use cases. In addition, the use cases confirm the coexistence of various Connector implementations with different functional characteristics that nevertheless all adhere to the principles of the reference architecture model.

The prerequisite here is conformity to the IDS information model. This criterion is being fulfilled in all cases.

This analysis also shows the step-by-step development or implementation of important functionalities, since usage control technologies are neither in use nor planned for all cases.

In addition, the need of action with respect to implementation and use of the app store and broker components becomes apparent.

Table 8.3 additionally shows to what extent the strategic requirements for the IDS architecture are a reality in the six use cases.

This analysis demonstrates that the focus in the use cases during the initial phases of the initiative lies on the standardized interface, which is in use in all of the examples. The requirement for decentralized data storage is also being fulfilled in nearly all of the use cases.

Table 8.2 Use of software prototypes in the use cases

		K1				K2	K3	K4	K5	K6
		Connector				Usage Control	Information Model	App Store	Base Broker	Identity Provider
		Base Connector	Trusted Connector	Sensor Connector	Embedded Connector					
F1	Collaborative SCM	X	(X)			X	X			
F2	Transparency in supply chains	X		X	X		X		(X)	
F3	Data trusteeship for industrial data		X			X	(X)		(X)	X
F4	Digital networking of manufacturing lines			X		(X)	X		X	
F5	Product lifecycle management in the business ecosystem	X				(X)	X		(X)	
F6	Agile networking in value chains	X				(X)	X	(X)	(X)	(X)

Key: X – prototype in use; (X) – use planned.

Specific requirements – such as those for a secure data supply chain and multiple operating environments – feature in those use cases where these specifications are relevant.

Areas that have been addressed only to a limited extent in the project's initial phase are the terms of use (only currently being utilized in F1) and the possibility to provide data apps. The latter is no surprise since an app ecosystem can only develop with the increasing dissemination of the initiative. The same applies to data linking: this requirement will increasingly move to the fore of implementation as use case scenarios become more complex.

The requirement to be able to attach terms of use to the data will become increasingly significant during future use case implementation phases when the basic communication is able to be conducted via the IDS and when increasingly sensitive data elements are being shared.

Table 8.3 Fulfillment of requirements in the use cases

		K1	K2	K3	K4	K5	K6
		Collaborative SCM	Transparency in supply chains	Data trusteeship for industrial data	Digital networking of production lines	Product lifecycle management in the business ecosystem	Agile networking in the value chain
A1	Terms of use for data	X		(X)		(X)	(X)
A2	Secure data supply chain		X				X
A3	Simple data linking	(X)	(X)		X	X	(X)
A4	Decentralized data storage	X	X	X		X	X
A5	Multiple operating environments		X		X		(X)
A6	Standardized interface	X	X	X	X	X	X
A7	Certification			X			
A8	Data apps and app store	(X)			X		(X)

Key: X – prototype in use; (X) – use planned.

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